Theoretical Handbook: IYPT 2027 — Pinhole Sunglasses

Complete Mathematical Model, Formula Reference, and Design Methodology

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Scope note. Formulas that carry the physics of this problem get full derivation, meaning, assumptions, validity, and limitations. Simple definitions get one compressed line. Materials/manufacturing content is numeric and load-bearing but not expanded into padding chapters — see Chapters 15–18.

PART I — Problem Framing

Chapter 1. Problem Statement

Official statement (IYPT 2027): "Investigate the optimal arrangement of pinholes in an opaque film to produce sunglasses that reduce light intensity and help correct short-sightedness."

Design variables: hole diameter a_{hole}; hole count N; pitch s; packing pattern; film thickness t; film material; vertex distance d_{gap}.

Objectives:

- **O1 (R1):** minimize retinal blur radius for a myopic eye viewing distant objects.
- **O2 (R2):** achieve a specified light-attenuation target.

Constraints: no ghosting (R3); acceptable field of view / gaze-angle performance (R4/R5); manufacturability (C3); experimental validation (R6); translation of blur into an actual visual-acuity claim (R7).

Structural fact (established in Chapter 9): O1 depends only on a_{hole}; O2 depends on N, a_{hole}, and pattern via open-area fraction. These are only weakly coupled — solve a_{hole} first, then N/pattern subject to constraints.

PART II — Optical System and Eye Model

Chapter 2. Optical System — Complete Path

Sun/object → pinhole array → air gap (d_{gap}) → cornea → aqueous humour → pupil → lens → vitreous humour –

The cornea, aqueous, crystalline lens, and vitreous are collapsed into a single **reduced eye**: one equivalent thin lens of focal length f, imaging onto a plane at distance v_r. Real refractive indices ($n_{cornea} \approx 1.376$, $n_{aqueous} \approx 1.336$, $n_{lens} \approx 1.386$–1.406, $n_{vitreous} \approx 1.336$) are already absorbed into f and are not carried separately through later formulas — a standard, stated simplification, not an oversight.

Chapter 3. Human Eye Model

- **Reduced eye:** $v_r \approx 17$ mm (convention used consistently throughout).
- **Diopter:** $D = 1/f$ (meters).
- **Myopia parametrization**, given magnitude M (diopters):

$$f = \frac{1}{D_{emmetropic} + M}, \qquad D_{emmetropic} \approx \frac{1}{v_r}$$

  Worked example used throughout Chapters 5–9: $v_r = 17$ mm $\Rightarrow D_{emmetropic} \approx 58.8$ D; taking $f = 16.7$ mm $\Rightarrow 1/f \approx 59.9$ D $\Rightarrow M \approx 1.1$ D (mild myopia).
- **Circle of confusion:** identical to r_{geo}, Chapter 5.
- **Depth of focus:** axial range over which r_{geo} stays under a tolerance — see Chapter 5's closing formula.
- **Retinal geometry:** treated as flat near the visual axis; this assumption is weakest in Chapter 13 (wide-field claims).

PART III — Wave Optics Foundations

Chapter 4. Wave Optics Hierarchy

Maxwell's equations → scalar wave equation → Helmholtz equation ($\nabla^2 U + k^2 U = 0$) → Huygens–Fresnel → Kirc

The Angular Spectrum Method (Chapter 7) is the **exact, non-paraxial** numerical implementation of Rayleigh–Sommerfeld propagation used as the production tool in this project; Fresnel and Fraunhofer closed forms are used only as validation limits, not as production formulas.

PART IV — Aperture Discovery

What diameter should every pinhole be?

Chapter 5. Geometrical Blur

5.1 — Thin lens equation

$$\frac{1}{v_f} = \frac{1}{f} - \frac{1}{u}$$

Variables: v_f, f, u in meters, real-is-positive convention. *Assumptions:* paraxial, thin lens. *Validity:* breaks down for non-paraxial rays (spherical aberration) — acceptable given the near-axis restriction used throughout. *Use:* fixes v_f for Formula 5.2.

5.2 — Geometric blur radius

$$r_{geo} = \frac{a_{stop}}{2} \cdot \frac{|v_r - v_f|}{v_f}$$

Derivation: similar triangles — a marginal ray converging to $(v_f, 0)$ diverges linearly past focus; height at v_r scales by the distance ratio past focus. *Validity:* good only when diffraction is negligible; superseded by the full ASM pipeline (Chapter 8) near the actual optimum. *Limitations:* retained here as the deprecated/pedagogical formula and the basis of the RSS comparison curve, not as the production value.

Depth of focus (inverting 5.2 for a tolerance r_{tol}):

$$\Delta v_f = \frac{2\, r_{tol}\, v_f}{a_{stop}}$$

Smaller a_{stop} gives larger depth of focus for the same tolerance — the actual mechanism claim for pinhole glasses, stated formally.

Chapter 6. Diffraction

6.1 — Airy intensity pattern

$$I(\theta) = I_0 \left[\frac{2J_1(ka \sin\theta)}{ka \sin\theta} \right]^2, \qquad k = \frac{2\pi}{\lambda}$$

Assumptions: far field (Fraunhofer), circular aperture, monochromatic. *Validity:* only where the far-field condition holds (Formula 6.4). *Use:* defines the ideal diffraction-limited spot; validation target for the numerical pipeline.

6.2 — First null radius

$$r_{diff} = 1.22\, \frac{\lambda v_r}{a}$$

Limitations: a null radius, not an energy-based size — see 6.3.

6.3 — Encircled energy

$$EE(x) = 1 - J_0^2(x) - J_1^2(x), \qquad x = \frac{\pi a r}{\lambda v_r}$$

At the first null ($x = 3.8317$): $EE \approx 0.838$ — the null-radius metric misses $\sim 16\%$ of the energy. Design metric used throughout this project: $r_{EE}(P)$, the radius enclosing energy fraction P (report $P = 0.5, 0.8, 0.9$).

6.4 — Fresnel number (regime selector)

$$N_F = \frac{a^2}{\lambda z}$$

$N_F \gg 1 \Rightarrow$ near field; $N_F \ll 1 \Rightarrow$ far field. Used as a validation/sanity gate around the numerical pipeline, not as a live branching rule.

MTF/OTF:

$$\mathrm{OTF}(f_x, f_y) = \mathcal{F}\{\mathrm{PSF}(x, y)\}, \qquad \mathrm{MTF} = |\mathrm{OTF}|$$

Extracted from experimental resolution-chart photographs and compared directly against the simulated PSF's own MTF — stronger evidence than comparing blur radii alone.

Chapter 7. Near-Field Propagation (Pinhole Film → Cornea)

7.1 — Angular Spectrum Method (exact scalar propagator)

$$\hat{U}(f_x, f_y, 0) = \mathcal{F}\{U(x, y, 0)\}$$

$$H(f_x, f_y, z) = \begin{cases} \exp\left(i\frac{2\pi}{\lambda} z\sqrt{1 - \lambda^2(f_x^2 + f_y^2)}\right) & f_x^2 + f_y^2 \le 1/\lambda^2 \\ 0 & \text{(evanescent, filtered)} \end{cases}$$

$$U(x, y, z) = \mathcal{F}^{-1}\{\hat{U}(f_x, f_y, 0) \cdot H(f_x, f_y, z)\}$$

Validity: exact for any z, near or far field — why this replaced closed-form estimates as the production method. *Limitations:* numerical — requires simultaneous Nyquist sampling of the smallest feature (hole radius) and adequate window size to avoid FFT wraparound. This dual-scale requirement was the confirmed source of an early validation discrepancy: error dropped from $0.30 \to 0.076 \to 0.019$ across $N = 1024/2048/4096$ at fixed physical window — confirming a sampling artifact, not a physics error.

7.2 (deprecated) — Quadrature effective-aperture estimate

$$a_{eff} \approx \sqrt{a_{hole}^2 + \left(\frac{2.44\,\lambda\, d_{gap}}{a_{hole}}\right)^2}$$

Cheap cross-check only; cannot represent the oscillatory near-field behavior confirmed by 7.3.

7.3 — On-axis Fresnel intensity

$$I(0, z) = 4I_0 \sin^2\left(\frac{\pi N_F}{2}\right)$$

Non-monotonic, oscillatory — direct evidence 7.2's smooth blending is structurally inadequate in this project's regime ($N_F \sim 10$–40).

Chapter 8. Combined Blur — Exact vs. Approximate

Deprecated: $r_{total} = \sqrt{r_{geo}^2 + r_{diff}^2}$ (RSS) — a photographic-engineering shortcut, an *approximation* to a convolution, not a derivation.

Exact (production) approach: propagate the actual defocused wavefront through the full chain using 7.1 twice (once over d_{gap}, once over v_r) with a lens-phase multiplication between:

8.1 — Thin lens phase transformation

$$U_{after}(x, y) = U_{before}(x, y) \cdot \exp\left(-i \frac{k}{2f}(x^2 + y^2)\right)$$

Validity: paraxial chirp approximation to a spherical wavefront — valid at this project's sub-mm to few-mm aperture scale.

Critical correction (documented failure mode, retained as a worked example): a naive implementation applies a single FFT immediately after 8.1, which is only correct if the observation plane is exactly the lens's own back focal plane. Since $v_r \neq f$ for a myopic eye, this silently evaluates the field at the wrong plane. **Fix:** replace that FFT with a second application of 7.1 over the physical distance v_r. This is what makes the "exact approach" true — without it, the pipeline was not modeling a defocused eye at all.

Validation performed: in the zero-defocus, zero-gap limit ($v_r = f$, $d_{gap} \to 0$), the corrected pipeline's encircled-energy curve converged to the closed-form Airy result (6.3) to within $\sim 2\%$ at $N = 4096$ over a 6 mm window ($dx \approx 1.5$ μm).

Chapter 9. Optimization

Objective: minimize $r_{EE}(P; a_{hole})$ over a_{hole}, reported for $P = 0.5, 0.8, 0.9$ — the optimum is P-dependent; a single number without a stated P misrepresents the result.

Closed-form (RSS-only) optimum, from $r_{total}^2 = C_1^2 a^2 + C_2^2/a^2$:

$$a^* = \sqrt{\frac{C_2}{C_1}}$$

Useful only as an analytic cross-check and search-bracket, not as the reported design value — the numerical optimum from the corrected pipeline was found to diverge from this in the project's own sweep, itself a reportable finding about RSS's bias direction.

Production result: report a table of $a^*(P)$ across $P \in \{0.5, 0.8, 0.9\}$ and across at least two myopia severities.

PART V — Pattern Discovery

How should the holes be arranged?

Chapter 10. Packing Geometry

10.1 — Square packing open-area fraction

$$\phi_{square} = \frac{\pi r_{hole}^2}{s^2}$$

10.2 — Hexagonal packing open-area fraction

$$\phi_{hex} = \frac{2\pi r_{hole}^2}{\sqrt{3}\, s^2}$$

$$\frac{\phi_{hex}}{\phi_{square}} = \frac{2}{\sqrt{3}} \approx 1.155$$

Hexagonal packing gives $\sim$ 15.5% higher transmission at equal spacing s — since the no-ghost constraint (Chapter 12) caps the *maximum* s and manufacturing sets a *minimum* s, hexagonal packing lets you hit a higher transmission target within the same allowable window.

Other patterns (radial, Fermat/golden-angle spiral, Voronoi, random): reduce periodic-ghosting salience but have no closed-form open-area or ghosting formula — quantify experimentally if pursued, don't assert.

Chapter 11. Brightness and Transmission

11.1 — Open-area transmission fraction

$$T = \frac{N \cdot (\pi a_{hole}^2/4)}{A_{total}} = \phi_{pattern}$$

11.2 — Transmitted intensity

$$I_{out} = T \cdot I_{in}$$

Optical density: $OD = -\log_{10} T$ (typical sunglasses target $T \approx$ 15–25%, $OD \approx$ 0.6–0.8).

Chapter 12. Ghost Images

12.1 — Retinal image shift between two holes

$$\Delta r_{retina} \approx s \cdot \frac{v_r}{d_{gap}}$$

12.2 — No-ghost condition

$$s < r_{EE}(P) \cdot \frac{d_{gap}}{v_r}$$

Combine with Chapter 10/11 to bound the achievable (N, s) region for a stated target T.

Interference: negligible for spatially incoherent extended-source viewing (daylight scenes) — an explicit, stated assumption, not a silent one; would not hold for a coherent point-source (e.g., laser) test.

Chapter 13. Field of View and Vignetting

13.1 — Pupil lateral displacement under eye rotation

$$\Delta(\varphi) \approx R_{rot} \sin\varphi$$

$R_{rot} \approx$ 13–14 mm (center-of-rotation to pupil-plane distance). *Validity:* restrict to normal gaze range ($|\varphi| \lesssim$ 20–30°), consistent with paraxial assumptions elsewhere.

13.2 — Overlap area of two circular apertures (pupil radius R_p, hole radius R_h, center separation Δ)

$$A_{overlap}(\Delta) = R_p^2 \cos^{-1}\left(\frac{\Delta^2 + R_p^2 - R_h^2}{2\Delta R_p}\right) + R_h^2 \cos^{-1}\left(\frac{\Delta^2 + R_h^2 - R_p^2}{2\Delta R_h}\right) - \frac{1}{2}\sqrt{(-\Delta + R_p + R_h)(\Delta + R_p - R_h)(\Delta -$$

valid for $|R_p - R_h| < \Delta < R_p + R_h$; $A = \pi \min(R_p, R_h)^2$ for $\Delta \le |R_p - R_h|$; $A = 0$ for $\Delta \ge R_p + R_h$.

13.3 — Acceptance angle Define φ_{50} where $A_{overlap}(\varphi)/A_{overlap}(0) = 0.5$, solved numerically from 13.2 combined with 13.1. For the full array, sum the per-hole overlap area across all N hole positions at each φ — peripheral holes extend the array's effective field of view beyond any single hole's acceptance angle.

Retinal curvature caveat: the flat-retina assumption is weakest here; state this explicitly rather than implying full peripheral-quality characterization.

Chapter 14. Manufacturing

- **Laser cutting:** minimum clean hole diameter $\approx$ 100–150 μm in thin polymer film; heat-affected-zone (HAZ) rim can reduce effective clear aperture by tens of microns — measure actual post-cut diameter, don't trust nominal.
- **Mechanical punching:** cleaner edges, no HAZ; tooling-limited to $\gtrsim$ 300 μm typically; better repeatability across many holes, less flexible.
- **Circularity tolerance:** even a few percent ellipticity measurably distorts the Airy pattern's symmetry (Formula 6.1) — measure before comparing model to experiment.

14.1 — Tunnel / acceptance-angle model (thickness effect)

$$\mathrm{FOV}_{hole} = 2\tan^{-1}\left(\frac{a_{hole}}{2t}\right)$$

Only matters if substrate thickness t is not negligible relative to a_{hole}. Example: $a_{hole} = 0.3$ mm, $t = 0.5$ mm (thick acrylic) $\Rightarrow$ $\mathrm{FOV}_{hole} \approx 31°$ (real constraint); same hole in $t = 50$ μm PET $\Rightarrow\approx 160°$ (negligible). Choosing thin film is a design decision informed by this formula.

Chapters 15–18. Material, Internal Reflections, Thermal (compressed)

Material	Use	Laser-cut edge	Min. hole Ø	Notes
Black PET/polyester film	Pinhole substrate	Good, some HAZ	$\sim$ 150 μm	Thin, flexible, cheap

Material	Use	Laser-cut edge	Min. hole Ø	Notes
Anodized/black aluminum foil	High-opacity thin layer	Very good	$\sim$ 100–150 μm	Best edge sharpness for small holes
Acrylic (PMMA), opaque-dyed	Rigid frame-mounted film	Moderate HAZ, discoloration	$\sim$ 200–300 μm	Thicker stock triggers 14.1's FOV constraint
Polycarbonate	Frame/lens body (not perforated layer)	N/A	—	Relevant to surrounding structure only
Stainless steel shim	Durable array	Excellent	$\sim$ 150–300 μm	Heavier, very repeatable

Internal reflections/thermal: at normal eyewear operating temperature range and sub-mm feature sizes, both effects are below the manufacturing tolerance and Chapter 9's sensitivity floor (typical linear expansion 10^{-5}–10^{-4} /°C applied to a $\sim$ 0.3 mm feature) — stated with a magnitude, not expanded into separate chapters.

PART VI — Simulation

Implemented in `pinhole_sim.py`: circular aperture generation, ASM propagation (7.1) with evanescent cutoff, lens phase transformation (8.1), the corrected two-stage pipeline (Chapter 8), numerical encircled-energy extraction (6.3), and a validation routine confirming convergence to the closed-form Airy result with increasing grid resolution. GPU acceleration not required at this problem's grid sizes; only worth pursuing for full 2D multi-hole array simulations.

PART VII — Experimental Verification

Artificial eye: converging lens ($f \approx 15$–20 mm) with sensor offset simulating a stated myopia diopter. USAF 1951 target imaged through interchangeable hole films; MTF extracted via slanted-edge analysis; transmitted intensity measured via lux meter/photodiode, calibrated against 11.1–11.2. Compare experimental MTF directly against the simulated pipeline's MTF — stronger validation than comparing single blur radii.

PART VIII — Multi-Objective Optimization / Final Design

Given a_{hole} from Chapter 9, pattern from Chapter 10 (hexagonal preferred), N and s solved jointly from 11.1 and 12.2 for a stated target T, FOV/vignetting from Chapter 13, manufacturing tolerance from Chapter 14: report a Pareto front of $r_{EE}(P)$ vs. T across candidate designs, with an explicit, justified tradeoff choice — not a single unqualified "optimal design."

Appendix A — Symbol Table

Symbol	Meaning	Unit
a_{hole}	physical pinhole diameter	m
a_{eff}	effective spot diameter after gap propagation (deprecated)	m
a_{stop}	effective aperture stop after pupil clipping	m
d_{gap}	pinhole film to cornea distance	m
v_r	lens-to-retina distance	m
v_f	lens-to-actual-focus distance	m
u	object distance	m
f	eye's pathological focal length	m
λ	wavelength	m
N_F	Fresnel number	—
r_{geo}	geometric blur radius	m
r_{diff}	diffraction first-null radius	m
$r_{EE}(P)$	radius enclosing energy fraction P	m
$EE(x)$	encircled energy function	—
N	hole count	—
s	hole pitch	m
T	open-area transmission fraction	—
M	myopia magnitude	D (m^{-1})
φ	eye gaze rotation angle	rad/deg
R_{rot}	center-of-rotation to pupil-plane distance	m
t	film substrate thickness	m

Appendix B — Formula Index

5.1 Thin lens · 5.2 Geometric blur radius · 6.1 Airy intensity · 6.2 First-null radius · 6.3 Encircled energy · 6.4 Fresnel number · 7.1 Angular Spectrum Method · 7.2 Quadrature effective-aperture (deprecated) · 7.3 On-axis Fresnel intensity · 8.1 Lens phase transformation · 10.1 Square packing fraction · 10.2 Hexagonal packing fraction · 11.1 Open-area transmission · 11.2 Transmitted intensity · 12.1 Retinal image shift · 12.2 No-ghost condition · 13.1 Pupil lateral displacement · 13.2 Circular-aperture overlap area · 13.3 Acceptance-angle definition · 14.1 Tunnel/FOV thickness model.

Appendix C — Simulation Pseudocode

```
function run_pipeline(a_hole, d_gap, v_r, f, wavelength, D_pupil, N, L):
    dx = L / N
    U0 = circular_aperture(N, dx, a_hole)
    U1 = ASM_propagate(U0, dx, wavelength, d_gap)      # Formula 7.1
    U2 = U1 * circular_aperture(N, dx, D_pupil)        # pupil clip
```

```
    U3 = U2 * lens_phase(N, dx, wavelength, f)         # Formula 8.1
    U4 = ASM_propagate(U3, dx, wavelength, v_r)        # Formula 7.1 again (corrected step)
    I_retina = |U4|^2
    r_bins, EE = radial_encircled_energy(I_retina, dx) # Formula 6.3, numerical
    return I_retina, r_bins, EE
```

Full working implementation (with validation routine and single-hole sweep) exists as `pinhole_sim.py` — attach alongside this handbook.